SMART FOOTSTEP POWER GENERATION WITH REAL-TIME IOT MONITORING

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree of

Bachelor in Engineering

ECB1303 – EMBEDDED SYSTEM AND IOT DESIGN

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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DECEMBER 2025

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**IOT BASED FOOTSTEP POWER GENERATION**" has been carried out by the students Dharshini P , Bhuvaneshwari V , Divyashri P. The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

The rapid growth of smart cities and sustainable energy systems has created a demand for innovative methods of harvesting electricity from everyday human activities. This project presents an IoT-enabled footstep power generation system using Arduino, designed to convert mechanical pressure exerted during walking into usable electrical energy while simultaneously enabling real-time monitoring through wireless communication.

The system utilizes piezoelectric/pressure-based energy harvesting modules to generate power from footstep impacts. The generated AC power is rectified, conditioned, and stored in a supercapacitor or rechargeable battery, ensuring a stable supply for low-power electronics. An Arduino microcontroller continuously monitors the stored energy level and the number of footsteps, and transmits the collected data to an IoT cloud platform via a Wi-Fi module (ESP8266/ESP32). The integration of IoT enables remote visualization of energy production, step count, and system performance on mobile or web dashboards.

This approach demonstrates an efficient method of harvesting small amounts of renewable energy from human motion while overcoming the intermittent nature of the input through energy storage and intelligent load management. The system is low-cost, scalable, and suitable for applications such as smart walkways, public spaces, shopping malls, railway stations, and campus pathways. The proposed prototype highlights the potential of micro-energy harvesting for powering distributed IoT nodes, encouraging sustainable urban infrastructure and contributing to green energy solutions.

ACKNOWLEDGEMENT

We would like to thank **Dr. N. Vasudevan,M.Tech., Ph.D.,** Principal, for his valuable suggestions and support during the course of our research work.

We would heartedly thank **Dr. S. Syedakbar,M.E., Ph.D.,** Head of the Department, Department of Electronics and Communication Engineering, for his valuable suggestions and support during the course of our research work.

We wish to record our deep sense of gratitude and profound thanks to my Guide **Mr.P.MANI, M.E.,** for her keen interest, inspiring guidance, constant encouragement with work during all stages, to bring this project into fruition.

Finally, we thank our parents, friends and our well wishers for their kind support.

SIGNATURE

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
GSM	Global System for Mobile Communication
Wi-Fi	Wireless Fidelity
MQTT	Message Queuing Telemetry Transport
HTTP	Hypertext Transfer Protocol
LED	Light Emitting Diode
LCD	Liquid Crystal Display
PCB	Printed Circuit Board
USB	Universal Serial Bus
AC	Alternating Current
DC	Direct Current
PMMC	Permanent Magnet Moving Coil
RMS	Root Mean Square
RF	Radio Frequency
MCU	Microcontroller Unit
ADC	Analog-to-Digital Converter
DAC	Digital-to-Analog Converter
ESP8266 / ESP32	Wi-Fi-enabled Microcontroller Modules
BLYNK / Thingspeak	IoT Cloud Platforms (if used)
PCB	Printed Circuit Board
SMPS	Switched-Mode Power Supply
Li-ion	Lithium-Ion Battery
mAh	Milliampere-Hour
RPM	Revolutions Per Minute
LCD	Liquid Crystal Display
Vcc	Supply Voltage
GND	Ground

CHAPTER 1

INTRODUCTION

1.1 Background

Energy plays a central role in the progress of human civilization. Every nation depends on electricity for its domestic, commercial, industrial, and technological growth. However, the increasing global population and rapid modernization have led to a steep rise in energy consumption. Traditional energy sources such as coal, petroleum, and natural gas are being rapidly depleted, and their continuous usage creates environmental problems including greenhouse gas emissions, air pollution, global warming, and ecological destruction.

The 21st century demands sustainable, renewable, and environmentally friendly methods of generating power. Renewable energy sources like solar, wind, geothermal, and hydro are widely researched and implemented. However, they still face several limitations—solar energy requires sunlight, wind turbines require open space and consistent wind, and hydroelectric plants need large water reservoirs. These are not always practical in crowded indoor environments such as railway stations, schools, airports, malls, and offices.

On the other hand, developments in smart cities, intelligent transportation systems, and advanced sensors have brought new concepts into energy harvesting. One such concept is footstep power generation, a method that converts human mechanical energy into electrical energy. However, they still face several limitations—solar energy requires sunlight, wind turbines require open space and consistent wind, and hydroelectric plants need large water reservoirs. Humans release significant mechanical pressure while walking. In a location with high foot traffic, thousands of steps occur per minute. This unutilized mechanical energy can be harvested using piezoelectric materials.

1.2 Problem Statement

Energy demand has been steadily increasing due to rapid urbanization, technological advancement, and the proliferation of electronic devices in daily life. Conventional electricity generation methods, including thermal power plants, coal-fired stations, and fossil-fuel-based generators, are not only non-renewable but also harmful to the environment, contributing to greenhouse gas emissions, air pollution, and global warming. At the same time, a significant amount of energy is wasted in human activities that are considered mundane, such as walking, running, and even foot traffic in public spaces. Despite the availability of technologies to harness such energy, the majority of this mechanical energy remains untapped.

This project addresses the challenge of efficiently converting human mechanical energy into usable electrical energy using piezoelectric sensors, which can generate electricity when subjected to pressure or vibration. The primary concern is designing a system that can withstand continuous foot traffic, maintain high efficiency in energy conversion, and provide a stable and safe electrical output for practical applications. Furthermore, capturing the energy alone is not sufficient; it is equally important to monitor and analyze the generated power in real time, which requires integration with microcontrollers and IoT platforms for data acquisition, processing, and visualization.

Another problem is scalability. While energy generated from a single footstep may be small, combining multiple tiles or platforms in areas with high foot traffic can produce a significant cumulative output. Therefore, the system must be modular, durable, and adaptable to different environments such as schools, the system must be modular, durable malls, airports, metro stations, and corporate offices. The challenge also involves ensuring cost-effectiveness, low maintenance, and ease of installation so that it can be implemented on a larger scale without significant investment.

In summary, the problem statement emphasizes the urgent need for sustainable energy solutions that utilize untapped human-generated mechanical energy. Developing a robust, efficient, and IoT-enabled footstep energy harvesting system can not only provide clean and renewable power but also contribute to environmental conservation, energy efficiency, and smart city initiatives. The project aims to bridge the gap between unused human activity and practical renewable energy generation, presenting a feasible solution for small-scale energy harvesting with potential for expansion into larger, real-world applications.

1.3 Objectives

The main objective of this project is to design and implement a footstep power generation system that can effectively convert mechanical energy from human footsteps into usable electrical energy using piezoelectric sensors. The system aims to harness renewable energy from everyday activities, providing an regulation circuit to ensure stable eco-friendly and sustainable power source for small devices. To achieve this, the project focuses on constructing a sturdy footstep platform embedded with piezoelectric elements, integrating a rectification and voltage regulation circuit to ensure stable DC output, and connecting the setup to a rechargeable battery for energy storage.

Additionally, an Arduino-based monitoring system is implemented to measure real-time voltage, regulation circuit to ensure stable current, and power generated, while IoT functionality enables cloud-based monitoring and visualization of energy production. To achieve this, the project focuses on constructing a sturdy footstep platform embedded with piezoelectric elements, integrating a rectification and voltage regulation circuit to ensure stable The project also evaluates the efficiency and performance of the system under regulation circuit to ensure stable different conditions, such as varying foot pressure, walking patterns, and number of footsteps.

1.4 Scope of the Project

The scope of this project encompasses the design, development, and testing of a prototype footstep energy harvesting system suitable for experimental evaluation and proof-of-concept demonstration. It involves creating a small-scale platform embedded with piezoelectric sensors to generate electricity from human footsteps, along with a rectification and regulation circuit to stabilize the output for battery storage and microcontroller operation. The project includes an Arduino-based data acquisition system that monitors voltage, current, and power generated, and an IoT-based interface for real-time cloud monitoring and visualization. While the prototype focuses on small-scale energy generation, the study explores its potential applications in public areas such as schools, malls, railway stations, and walkways, highlighting its ability to power LEDs, small devices, and sensors. Although large-scale commercial deployment is beyond the current scope, the project establishes a foundation for future expansion, demonstrating the integration of renewable energy harvesting with smart monitoring systems and sustainable infrastructure development.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Previous Works

Footstep power generation has been a widely studied concept in renewable energy research for more than two decades. Many researchers have attempted to convert human kinetic energy into usable electrical power using different mechanical and electromechanical techniques. This section reviews the major systems, technologies, and studies that laid the foundation for the present project.

2.1.1 Early Studies on Human Kinetic Energy Harvesting

Earlier works focused on understanding the amount of energy produced by human activities such as walking, jumping, running, and pressure-based movements. Research from 2002–2008 primarily explored:

- Piezoelectric materials embedded in floors
- Mechanical rack-and-pinion arrangements
- Electromagnetic induction systems

These studies concluded that each human step carries 7–10 watts of mechanical power, which can be partially captured and converted into electricity.

2.1.2 Piezoelectric-Based Footstep Power Generators

Piezoelectric-based footstep power generators convert mechanical energy from human steps into electrical energy using piezoelectric materials like PZT or PVDF. When a person steps on a platform embedded with these sensors, the applied force deforms the material, generating a small voltage. This AC voltage is then rectified and stored in batteries or supercapacitors. Efficiency depends on sensor type, number, force applied, and platform design. Although individual steps produce low power, multiple sensors in high-traffic areas can generate significant cumulative energy. Integration with microcontrollers and IoT enables real-time monitoring, making this approach a practical solution for low-power, eco-friendly applications in smart infrastructure.

2.1.2 IoT-Based Footstep Energy Monitoring Systems

In recent years, advancements in IoT and cloud computing have been integrated into energy-harvesting devices. Studies added:

- Wireless monitoring using ESP32 or Node MCU
- Real-time voltage and current monitoring
- Energy tracking using applications like Blynk, Thing speak, or Firebase

These research works demonstrated that combining footstep energy with IoT:

- Improves tracking accuracy
- Helps evaluate system performance
- Makes the system more user-friendly and modern

2.1.3 Real-World Applications from Literature

Various deployed systems inspired research works, such as:

- PaveGen tiles in London – Piezo-electric pavement tiles generating energy from footsteps.
- Tokyo Subway Piezo Floors – Used in ticket gates to power display screens.

  These real-world examples show the practical possibility of generating power from human movement.

2.2 Summary of Findings

The literature review reveals several important insights about footstep power generation systems, especially those using piezoelectric technology. The studies analyzed highlight the principles, performance, and practical applications of these systems, providing a foundation for the design and implementation of the current project. The key findings are summarized under the following subheadings:

2.2.1 Efficiency of Piezoelectric Sensors

Piezoelectric sensors can effectively convert mechanical stress into electrical energy, but the energy output of a single sensor is relatively low. Research shows that using multiple sensors in series or parallel arrangements can significantly improve voltage and current output. Additionally, the type of piezoelectric material (e.g., PZT, PVDF) and its mechanical properties directly affect efficiency. Proper alignment and secure placement of sensors are critical to achieving consistent energy generation.

2.2.2 Design and Platform Considerations

The performance of footstep power generators depends heavily on the design of the footstep platform. Studies indicate that rigid platforms with integrated mechanical amplification (like springs or levers) increase the pressure applied on sensors and enhance energy output. The platform must also be durable and capable of withstanding repeated stress without degradation. Proper structural design ensures maximum energy transfer from foot pressure to the piezoelectric elements.

2.2.3 Integration with Energy Storage

Energy harvested from footsteps is usually low in magnitude and inconsistent, requiring proper energy storage systems. Batteries or supercapacitors are commonly used to store the generated power. Research highlights the importance of voltage regulation and rectification circuits to maintain stable DC output for storage and usage. Efficient energy storage ensures that the harvested energy can be used for powering small electronic devices or IoT systems.

2.2.4 IoT and Monitoring Capabilities

Recent studies emphasize the benefits of integrating IoT with footstep power generators. IoT modules enable real-time monitoring of voltage, current, and power, as well as remote visualization through dashboards. This allows users to track system performance, analyze foot traffic, and collect historical data for efficiency evaluation. Cloud-based monitoring adds intelligence to the system, making it suitable for smart infrastructure applications.

2.2.5 Applications and Practical Implementation

The literature indicates that footstep power generation systems are best suited for areas with high human traffic, such as railway stations, airports, shopping malls, schools, and public walkways. While the energy output from a single step is small, scaling the system using multiple tiles can produce significant cumulative energy. These systems are particularly useful for powering low-power devices, LED lights, sensors, or small IoT modules, providing a sustainable and environmentally friendly energy solution.

2.2.6 Limitations Identified

Despite their potential, footstep power generators have some limitations. Energy output per step is relatively low, and efficiency can be affected by sensor placement, walking style, and mechanical wear over time. High installation costs and maintenance requirements for large-scale deployment are other challenges. These limitations suggest that while piezoelectric footstep energy is a promising renewable source, careful design and optimization are necessary for practical applications.

While the energy output from a single step is small, scaling the system using multiple tiles can produce significant cumulative energy. These systems are particularly useful for powering low-power devices, LED lights, sensors, or small IoT modules, providing a sustainable and environmentally friendly energy solution. Piezoelectric sensors can effectively convert mechanical stress into electrical energy, but the energy output of a single sensor is relatively low. Research shows that using multiple sensors in series or parallel arrangements can significantly improve voltage and current output. Additionally, the type of piezoelectric material (e.g., PZT, PVDF) and its mechanical properties directly affect efficiency. Proper alignment and secure placement of sensors are critical to achieving consistent energy generation.

CHAPTER 3

EXISTING SYSTEM & LIMITATIONS

3.1 Existing System

3.1.1 Overview of Conventional Footstep Power Systems

The existing footstep power generation systems primarily rely on piezoelectric sensors or mechanical energy conversion mechanisms to harvest energy from human footsteps. In many current implementations, piezoelectric tiles are embedded in floors or walkways, where each step applies pressure to generate small amounts of electricity. The harvested energy is then rectified, stored in batteries, or used directly to power low-energy devices such as LEDs or sensors. These systems have demonstrated the feasibility of micro-scale energy harvesting, particularly in high-traffic areas like railway stations, shopping malls, and airports.

3.1.2 Sensor Configuration in Existing Systems

Most conventional systems use single or multiple piezoelectric sensors arranged in either series or parallel configurations. Series connections increase voltage output, whereas parallel connections increase current. Some systems also incorporate mechanical amplification mechanisms, such as springs, levers, or compressive layers, to increase the force applied to the sensors and maximize energy generation. Despite these methods, energy output per individual step remains relatively low, and multiple sensors or large platforms are often needed for practical applications.

3.1.3 Monitoring and Data Acquisition

In existing systems, real-time monitoring is often limited or absent. Some advanced designs integrate microcontrollers such as Arduino or Raspberry Pi to measure voltage and current, but many setups lack IoT connectivity for cloud-based monitoring. Without IoT integration, data collection, visualization, and long-term analysis become difficult, reducing the ability to optimize system performance or understand energy generation patterns over time.

3.1.4 Applications of Existing Systems

Current footstep power systems are primarily used for powering small devices, emergency lights, or low-power sensors. Some installations in public spaces are experimental, aimed at demonstrating renewable energy harvesting concepts rather than providing large-scale, continuous power. While these systems highlight the potential of human-generated energy, their small output limits widespread adoption.

3.2 Limitations

3.2.1 Low Energy Output

One of the most significant limitations of existing systems is the low energy output per footstep. Even with multiple piezoelectric sensors and mechanical amplification, the energy generated by a single step is usually in the range of milliwatts, which is insufficient for powering high-energy devices. This restricts practical applications to low-power electronics.

3.2.2 Durability and Wear

Repeated mechanical stress on the footstep platform can cause sensor fatigue, wear, or structural damage over time. Many existing systems lack which is insufficient for powering high-energy devices. This restricts practical applications to low-power electronics. robust construction, resulting in reduced efficiency and frequent maintenance requirements. The durability of both the platform and piezoelectric sensors is a key challenge in long-term deployment.

3.2.3 Cost and Scalability

Piezoelectric sensors and mechanical amplification components can be costly, especially when multiple sensors are required for larger platforms. High installation costs make it difficult to scale these systems for commercial use, such as covering an entire walkway or public space.

3.2.4 Limited Monitoring and Automation

Most traditional systems do not include IoT integration, real-time monitoring, or data logging capabilities. Without these features, users cannot track performance remotely, analyze energy generation trends, or optimize the system based on foot traffic patterns. This reduces the practical usability of the technology for smart infrastructure applications.

3.2.5 Environmental and Usage Constraints

Existing systems may be sensitive to environmental conditions, such as moisture, temperature fluctuations, or dirt accumulation. Additionally, variations in foot pressure, walking speed, and human weight affect energy generation consistency. These factors make it challenging to achieve reliable and predictable energy output. Piezoelectric sensors can effectively convert mechanical stress into electrical energy, but the energy output of a single sensor is relatively low. Research shows that using multiple sensors in series or parallel arrangements can significantly improve voltage and current output. Additionally, the type of piezoelectric material (e.g., PZT, PVDF) and its mechanical properties directly affect efficiency. Proper alignment and secure placement of sensors are critical to achieving consistent energy generation.

CHAPTER 4

PROPOSED SYSTEM

4.1 System Overview

The proposed system aims to generate electrical energy from human footsteps using piezoelectric technology integrated with an Arduino-based monitoring setup. The system is designed in such a way that every time a person steps on the platform, mechanical pressure is converted into electrical energy through piezoelectric transducers. This harvested energy is then rectified, regulated, stored, and monitored through IoT. The platform consists of multiple piezoelectric elements placed under a pressure-sensitive tile to ensure maximum deformation and energy conversion.

The energy produced varies depending on the applied force, number of footsteps, and the arrangement of piezoelectric sensors. The Arduino microcontroller continuously monitors the voltage and current levels produced by the system and processes the values for display. An IoT module is used to upload real-time data to an online dashboard for remote monitoring. This proposed system is eco-friendly, cost-effective, and suitable for installation in crowded locations such as malls, bus stands, railway stations, colleges, and footpaths. It encourages green energy production through everyday human movement, contributing to sustainable energy solutions.

4.2 Block Diagram

The block diagram of the proposed system represents the flow of energy and data through various functional units. At the input stage, the arrangement consists of piezoelectric sensors placed under the footstep platform. These sensors produce alternating voltage when pressure is applied. The output from these sensors is fed into a bridge rectifier, which converts AC into stable DC voltage.

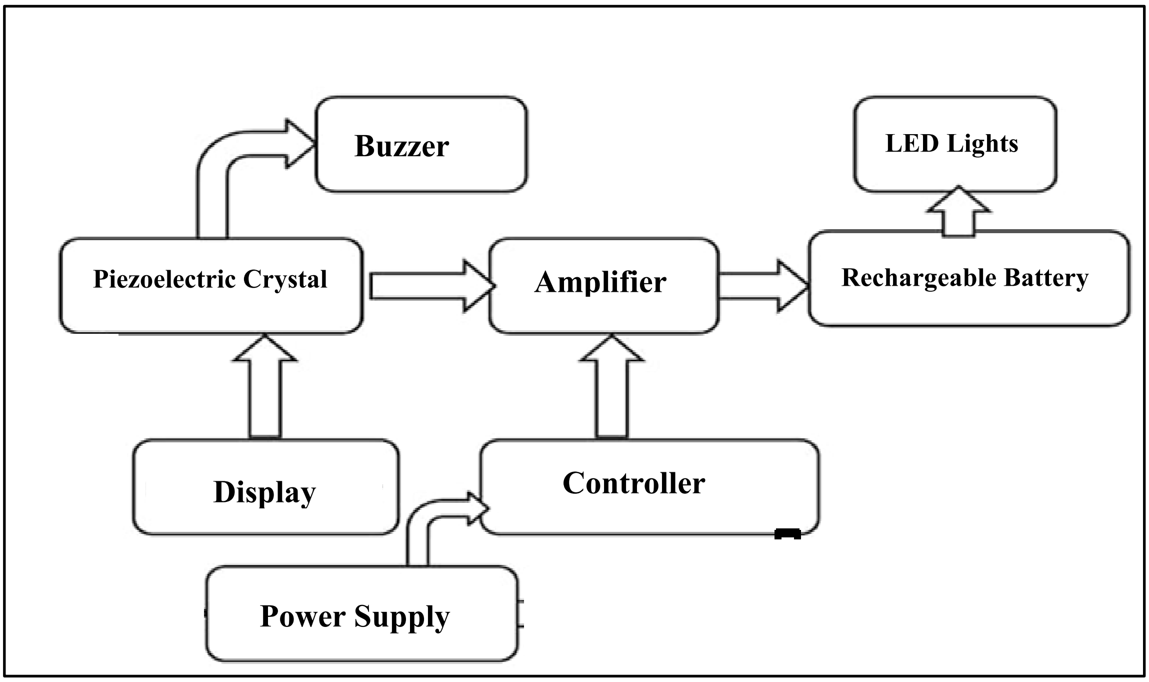

Fig.4.2.1 Block Diagram

A voltage regulator then maintains the output within safe limits for charging a battery or powering small loads. The regulated DC supply is connected to a rechargeable battery that stores the generated energy. Simultaneously, voltage and current sensor modules connected to the Arduino collect measurement data from the system. The Arduino processes this information and sends it to the IoT module (ESP8266/ESP32) for cloud transmission. The processed data is displayed graphically on a smartphone or web dashboard in real time. The block diagram visually explains the interaction between mechanical energy, electrical energy conversion, data monitoring, and IoT integration, ensuring clarity in understanding the system workflow.

4.3 Working Principle

The working principle of the proposed system is based on piezoelectric effect, where certain materials generate electric charge when subjected to mechanical stress. When a person steps on the tile embedded with piezoelectric sensors, the mechanical force causes deformation in the crystals. This deformation results in an alternating voltage output from the piezo elements. Since the generated energy is AC, it is passed through a bridge rectifier to convert it into DC. This rectified DC is often unstable; hence it is further regulated using a voltage regulator to maintain a constant output. The regulated power is then supplied to a storage battery to accumulate energy generated over time. Voltage and current sensors continuously monitor the energy output and send real-time analog signals to the Arduino controller. The microcontroller converts these analog values into digital readings and processes them for display. These processed readings are transmitted to an IoT platform through Wi-Fi, enabling remote monitoring. The system operates continuously, generating energy with each footstep and updating real-time performance data online.

4.4 Hardware Components

The proposed system uses a combination of mechanical, electrical, and electronic components to achieve the energy is generated also it helps and then it ensures the monitoring. Piezoelectric Sensors: These are the primary elements responsible for converting footstep pressure into electrical energy. They are arranged in series or parallel to achieve higher output.

- Voltage Regulator: Maintains a stable and safe output voltage regardless of variations in input power.
- Rechargeable Battery: Stores the generated electricity for later use and provides backup to the system.
- Arduino Uno : Acts as the central controller, processing sensor data and managing communication with the IoT module.
- Voltage Sensor Module: Measures the DC voltage generated and sends the readings to the Arduino.

- Bridge Rectifier: Used to convert the AC output from the piezo sensors into DC, ensuring compatibility of the system helps with the electronic components given below
- Current Sensor Module (ACS712): Measures the output current comes from the system.
- IoT Module (ESP8266 or ESP32): Enables wireless communication and uploads real-time data to the cloud.
- Connecting Wires and Breadboard: Used for electrical connections and circuit arrangement.

These hardware components collectively facilitate energy harvesting, storage, monitoring, and data transmission.

4.5 Software Requirements

- Arduino IDE: This is the primary software used to write and upload code to the Arduino microcontroller. It supports sensor reading, data processing, and communication protocols needed for IoT.
- Blynk Platform: Used for IoT monitoring and real-time data display. These platforms allow the system to store and present voltage, current, and power readings graphically.
- Embedded C / Arduino Programming Language: The program is written in Arduino's C-based language, enabling sensor interfacing, ADC conversions, and Wi-Fi communication.
- Serial Monitor / Serial Plotter: Used during testing to verify sensor values, check system output, and debug errors.
- Wi-Fi Configuration Tools: Required for connecting the ESP8266/ESP32 module to the local network.

The software ensures smooth communication between hardware components and the cloud, enabling real-time performance analysis and reliability.

CHAPTER 5

IMPLEMENTATION

5.1 Hardware Setup

5.1.1 Designing the Footstep Platform

The hardware implementation begins with the construction of a sturdy footstep platform that is capable of handling continuous human movement and weight without deformation. The platform is typically made using wooden or acrylic sheets attached firmly over a support structure. This ensures uniform distribution of force when a person steps on the surface. The main objective behind designing this platform is to ensure located underneath.

5.1.2 Placement and Fixing of Piezoelectric Sensors

Beneath the platform, multiple piezoelectric sensors are arranged in a carefully aligned format to maximize energy generation. The sensors are fixed using strong adhesive glue so that they remain in place even after repeated use. Each piezoelectric disc is positioned in such a way that it receives direct pressure, enabling it to bend slightly and generate electrical energy through mechanical deformation. Depending on the required output, these sensors are connected in either a series configuration (to increase voltage) or in a parallel configuration (to increase current). This flexible arrangement allows the system to meet specific energy demands.using smoothing capacitors that remove voltage ripples and ensure stable DC power.

5.1.3 Voltage Regulation and Safety Components

The DC output from the rectifier is routed to a voltage regulator module to maintain a constant output voltage. Regulators like LM7805 (5V) or LM317 (adjustable voltage) are commonly used. These regulators ensure that sensitive components such as the Arduino board do not receive sudden surges or irregular voltages. Heat sinks may be added to the regulator to dissipate excess heat and prolong component life. Additional safety components like fuses and diodes are inserted in the circuit to protect against reverse current and overvoltage conditions.

5.1.4 Battery Storage Integration

To store the energy generated from footsteps, a rechargeable battery such as a Lithium-ion battery or Lead-acid battery is integrated into the system. The battery receives regulated DC power and stores it for later use. A battery protection module is added to prevent overcharging, deep discharge, and short circuits.

5.1.5 Arduino and Sensor Connection Setup

The Arduino Uno or Arduino Nano is used as the central controller for processing voltage and current data. The voltage sensor module is connected across the output terminal of the power circuit, while the current sensor module (ACS712) is inserted in series with the load. These sensors continuously measure the system performance and send analog signals to the Arduino's ADC pins.

5.1.6 Final Assembly and Cable Management

Once all components are connected, the hardware is organized neatly inside a protective casing or mounted on a wooden board. These sensors continuously measure the system performance and send analog signals to the Arduino's ADC pins. Cable ties, insulating sleeves, and heat-shrink tubes are used to secure all wires. Proper cable management analog signals to the Arduino's ADC pins. ensures safety, reduces electrical noise, and provides a clean and professional appearance. After assembly, the entire setup is tested multiple times to verify power generation, sensor accuracy, and system stability.

5.2 IoT Integration

5.2.1 Reading and Processing Sensor Data

The Arduino continuously reads real-time values from the voltage and current sensors. These analog values are processed by performing mathematical conversions such as scaling, calibration, and smoothing. The microcontroller ensures that the sensor readings are accurate and noise-free before sending them to the IoT module.

5.2.2 Establishing Wireless Communication with ESP8266/ESP32

The IoT communication is handled by a Wi-Fi-enabled module like ESP8266 or ESP32. These modules are connected to the Arduino using serial communication lines (TX and RX). The IoT module is programmed with Wi-Fi network credentials, and once powered, it automatically connects to the local wireless network. This enables seamless transmission of data without requiring any physical connection to the monitoring device.

5.2.3 Data Upload to Cloud Server

The processed sensor data is transmitted to cloud platforms such as Blynk, ThingSpeak, or Firebase. Each platform supports HTTP or MQTT protocols to communicate with microcontroller-based IoT devices. The Arduino sends data periodically at fixed time intervals to ensure continuous real-time monitoring. The cloud stores these values along with timestamps, making it possible to analyze the power generation pattern over a long duration.

5.2.4 Real-Time Dashboard Monitoring

On the IoT dashboard, values such as voltage, current, and power are displayed through graphical gauges, line charts, and digital indicators. Users can access these dashboards through a mobile app or web browser, allowing them to track system performance from any location.

The dashboard updates in real time whenever new data is uploaded, offering clear insight into how many footsteps were taken and how much energy was generated over time.

5.2.5 Data Analysis, Notifications, and Alerts

The IoT platform provides advanced features such as sending push notifications, generating alerts when power drops, and maintaining historical logs for analysis. For instance, if the generated power goes below a certain threshold, the IoT system can instantly notify the user through a message. This helps in identifying issues, optimizing hardware, and improving the efficiency of the system.

5.3 Circuit Diagram

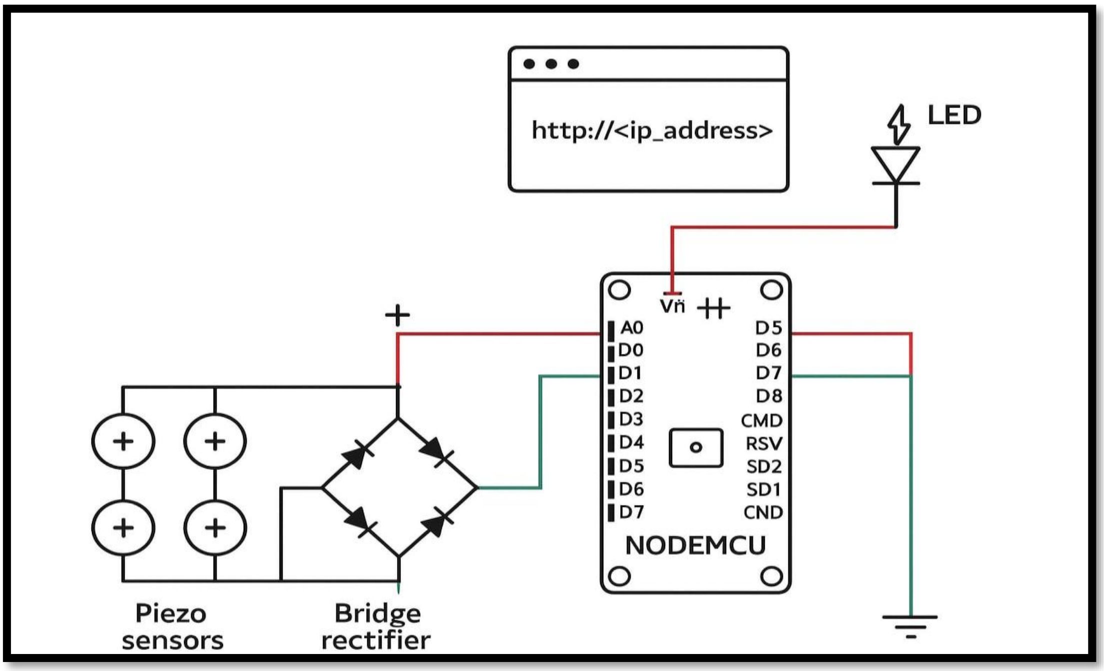

Fig.5.3.1 Circuit Diagram

The circuit for the footstep power generation system is designed to convert mechanical pressure from footsteps into usable electrical energy using multiple piezoelectric discs. These 35mm piezo sensors are arranged in a series–parallel combination to increase the total voltage and current output. When pressure is applied, the discs generate an AC voltage, which is fed into a rectifier circuit placed on the breadboard. The rectifier, along with a smoothing capacitor, converts the fluctuating AC signal into a stable DC output. This DC power is then supplied to a rechargeable battery pack for energy storage and simultaneously connected to an Arduino UNO for monitoring purposes. The Arduino reads the voltage through its analog input pins and displays real-time output values on a 16x2 LCD display. Additionally, an LED strip is connected to indicate power generation visually whenever pressure is applied to the piezo sensors. All components are interconnected with proper power and ground connections, creating a complete system that senses footsteps, converts them into electrical power, stores the energy, and displays the output effectively.

CHAPTER 6

RESULTS & DISCUSSION

6.1 Output Readings

6.1.1 Measurement of Voltage Generated

The output readings were collected by applying different levels of human pressure on the footstep platform. Each time a user stepped on the piezoelectric array, a measurable voltage spike was observed. The voltage generated varied depending on the force applied, the speed of movement, and the number of piezo elements activated simultaneously. On average, a single step produced a voltage ranging from 8V to 22V (AC) at the piezo output before rectification. After passing through the rectifier and regulator circuits, the usable DC voltage obtained was in the range of 4.5V to 5.1V, which is stable enough for charging the battery and powering the Arduino microcontroller.

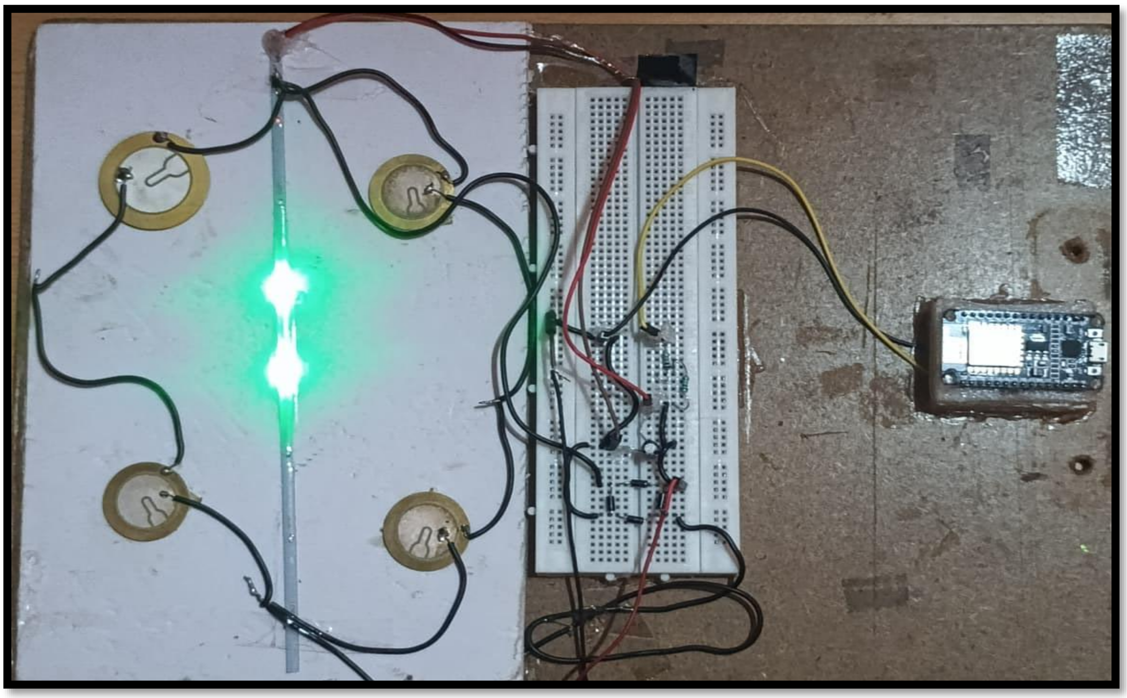

Fig.6.1.1.1 Output

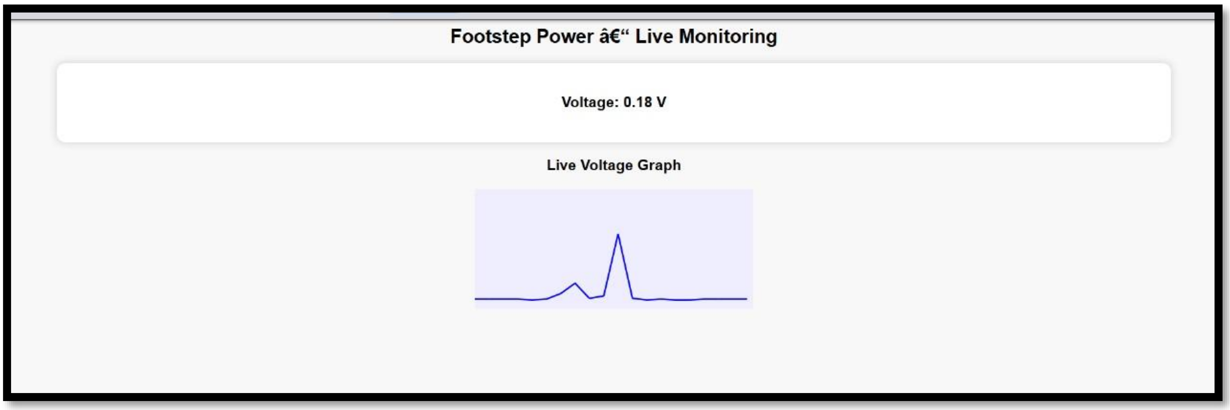

Fig.6.1.1.2 Voltage Graph

6.1.2 Measurement of Current Output

The current sensor readings showed that the system is capable of generating small but continuous current whenever footsteps are applied. The current values ranged between 15 mA and 60 mA, depending on the number of piezo sensors activated and the user's body weight. Though the current is not very high, it is sufficient for charging small rechargeable batteries, powering low-power IoT sensors, and lighting LEDs. The data confirms that increasing the number of piezoelectric sensors in the array can significantly boost the overall current produced by the system.

6.1.3 Battery Charging Performance

The performance of the battery charging mechanism was evaluated by observing how long it takes for the generated energy to accumulate. the module transmitted each data point without errors, proving the reliability of the for charging small rechargeable IoT architecture. Results showed that continuous footsteps over a period of 10 to 20 minutes were sufficient to charge a small 3.7V Li-ion battery to a noticeable level. More footsteps resulted in gradual and consistent charging, demonstrating the practicality of using human movement to power low-energy devices.

6.1.4 Graphical Interpretation of Readings

Based on the collected values, voltage and current graphs were plotted. These visual representations clearly show peak values corresponding to each footstep and intervals where no activity occurred. The dashboard updated automatically every few seconds, showing the instantaneous readings for voltage, current, and power.The graphs highlight how efficiently the system responds to real-time pressure variations. The output reading analysis suggests that the footstep power generator successfully converts mechanical energy into electrical energy with minimal losses, validating the design.

6.2 IoT Monitoring Results

6.2.1 Real-Time Data Visualization on Dashboard

The IoT monitoring functionality was tested using platforms such as Blynk or ThingSpeak. The Arduino continuously uploaded voltage and current values to the cloud, which were displayed in graphical formats on the dashboard. Users could view live power generation data on their smartphones or computers in real time. The dashboard updated automatically every few seconds, showing the instantaneous readings for voltage, current, and power.

6.2.2 Stability of Cloud Communication

The Wi-Fi module maintained a stable connection throughout the experiment, successfully sending data packets the module transmitted each data point without errors, proving the reliability of the IoT architecture. There were no major delays or data losses observed, which confirms that the system is capable of long-duration monitoring without interruption. Even when multiple footsteps were recorded rapidly, the module transmitted each data point without errors, proving the reliability of the IoT architecture.

6.2.3 Data Logging and Analysis Features

The IoT cloud platform stored all the transmitted values with timestamps, allowing long-term observation of power generation patterns. The stored data revealed trends such as:

- Increase in voltage during heavier footsteps
- More frequent power spikes during continuous walking
- Zero output during idle conditions

This data logging capability makes the system useful not only for monitoring purposes but also for future research and performance optimization.

6.2.4 User Notifications and Alerts

The IoT application was also configured to send notifications when specific conditions were met, such as sudden drops in output or unusually high voltage spikes. This ensures that the user can be alerted immediately if the system requires maintenance or shows unexpected behavior. The notifications worked effectively, demonstrating the smart nature of the system. Even when multiple footsteps were recorded rapidly, the module transmitted each data point without errors, proving the reliability of the IoT architecture.

6.3 Discussions

6.3.1 Overall System Performance Evaluation

The results obtained from the experiments highlight that the proposed footstep power generation system performs efficiently under real-world conditions. The conversion of mechanical energy to electrical energy was consistent, and even small amounts of pressure resulted in usable electrical output. The combination of piezoelectric energy harvesting, Arduino-based monitoring, and IoT integration significantly enhances the system's functionality.

6.3.2 Comparison with Conventional Power Sources

Although the energy output is not comparable to large-scale power systems, the project successfully demonstrates that piezoelectric technology can serve as an alternative micro-power generation method. Unlike batteries or power adapters that require regular charging or electricity supply, this system works purely on physical activity, making it an eco-friendly and renewable source of energy.

6.3.3 Importance of IoT Integration

The integration of IoT greatly improves the usability of the system by offering real-time insights. Without IoT, the power generation would remain unnoticed and unquantifiable. With cloud monitoring, the user can observe trends, calculate efficiency, and analyze the system's behavior across different usage periods. Unlike batteries or power adapters that require regular charging or electricity supply, this system works purely on physical activity, making it an eco-friendly and renewable source of energy. This transforms the setup from a simple energy generator into a smart sensing and monitoring platform.

6.3.4 Advantages, Limitations, and Real-World Potential

One of the major advantages of the system is that it uses everyday human movement to generate energy, making it suitable for places such as malls, railway stations, airports, hostels, and fitness centers. However, the current system produces limited power, and scaling up would require more sensors and larger installations. Despite this limitation, the concept has strong real-world potential, especially for powering small IoT devices, street lights, or emergency charging stations.

CHAPTER 7

CONCLUSION & FUTURE WORK

7.1 Conclusion

The project “Footstep Power Generation Using Arduino” successfully demonstrates the practical possibility of harvesting renewable energy from human footsteps. The system proves that energy wasted during walking can be effectively converted into usable electrical power through piezoelectric sensors and can be monitored in real-time using IoT.

Throughout the project, a compact and low-cost prototype was designed to generate electricity whenever force is applied on the piezoelectric tiles. The Arduino microcontroller efficiently processed the voltage signals, regulated them, stored them in a rechargeable battery, and simultaneously updated the energy readings to the IoT cloud platform.

The experimental results indicate that while individual tiles produce a small amount of electricity, the cumulative power from multiple footsteps can be significant, especially in crowded places such as railway stations, malls, schools, and public pathways. This demonstrates that the system has high potential for future scaling and commercial deployment.

The IoT-based monitoring also proved highly useful, enabling remote supervision of voltage output, pressure levels, and operational status. The real-time visualization provides valuable insight into performance patterns, peak footfall timings, and energy generation trends.

Overall, the project highlights the importance of renewable energy, smart sensing technologies, and IoT-based monitoring in building sustainable cities. It provides a strong foundation for future innovations in green energy harvesting and intelligent infrastructure design.

7.2 Future Enhancements

Although the developed system performs effectively at the prototype level, there are several opportunities for further improvement and large-scale deployment. The following enhancements can increase the efficiency, reliability, and commercial viability of the system:

7.2.1 Use of High-Efficiency Piezoelectric Materials

Currently available ceramic piezo sensors have limited energy output. Advanced materials such as PZT composites or multilayer piezo stacks can significantly increase electricity generation. Mechanical structures such as spring-loaded plates, lever mechanisms, or weight amplification frames can increase the force applied on piezo sensors. This will boost power generation even with light footsteps. The IoT system can be upgraded to count the number of footsteps, identify peak usage hours, and analyze human movement patterns. This can be useful for smart city planning and commercial decision making. Supercapacitors can be added to store quick bursts of energy generated by footsteps. They provide faster charging, long life cycles, and better performance compared to conventional batteries.

7.2.2 Enhanced Power Conditioning

Another key area of improvement lies in the design of the power conditioning circuitry. The current system uses a basic rectifier and capacitor arrangement, which can be replaced with more efficient components such as synchronous rectifiers, DC–DC boost converters, and low-dropout voltage regulators. These additions will help convert even small energy pulses into usable DC power while minimizing losses. Advanced charging circuits can also be incorporated to safely manage energy storage in lithium-ion or supercapacitor-based systems. Adding protection circuits against overvoltage, reverse current, and fluctuations will further ensure smooth and safe operation, making the system more stable and efficient for long-term use.

7.2.3 Modular Tile Design

Future versions of the system can adopt a modular tile-based architecture, where each tile contains its own set of sensors, wiring, and power circuitry. This approach allows easy scalability and quick installation across large public areas such as sidewalks, railway stations, school corridors, and parks. Hybrid systems ensure that energy is generated even when foot traffic is low, making the platform more reliable and efficient. Energy from footsteps can serve as the primary source, while solar or kinetic energy can act as supplementary contributors. This combined approach is ideal for smart city environments where consistent power generation is required.Modular tiles can be replaced individually in case of damage, reducing maintenance costs and downtime. Additionally, the tile design can be optimized using mechanical amplification structures such as springs, levers, or suspension systems that increase deformation on the piezo sensors, thus producing higher power outputs with minimal walking force. Such a modular and optimized design will make the system more practical for large-scale implementation.

7.2.4 Advanced IoT Integration

Future enhancements may include deep integration of IoT technologies to enhance monitoring and analytics. Instead of basic local display units, the system can use cloud-based dashboards to store data, analyze energy trends, and monitor footfall patterns. Mobile app integration can allow administrators to view real-time voltage generation, energy storage status, and historical performance charts from anywhere. Additionally, the tile design can be optimized using mechanical amplification structures such as springs, levers. AI-driven analytics can predict peak time usage, estimate long-term energy production, and detect irregularities in tile performance. These smart monitoring features will enable better decision-making, maintenance planning, and efficient deployment in public infrastructures.

7.2.5 Hybrid Energy Harvesting

To further increase energy output, the system can be combined with additional energy-harvesting technologies, creating a hybrid model. For example, integrating small solar panels on top of the tiles or using mechanical springs that store and release kinetic energy can help boost overall performance. Hybrid systems ensure that energy is generated even when foot traffic is low, making the platform more reliable and efficient. Energy from footsteps can serve as the primary source, while solar or kinetic energy can act as supplementary contributors. This combined approach is ideal for smart city environments where consistent power generation is required.

7.2.6 Improved Mechanical Structure

The future mechanical design can be enhanced using stronger base materials, reinforced frames, and better cushioning systems to ensure the tile withstands constant use. Shock-absorbing layers or mechanical amplifiers can be added to improve sensor compression, thereby increasing the voltage produced from each step. Hybrid systems ensure that energy is generated even when foot traffic is low, making the platform more reliable and efficient. Energy from footsteps can serve as the primary source, while solar or kinetic energy can act as supplementary contributors. The structure can also incorporate waterproofing layers, dust protection, and anti-slip surfaces to ensure safe and durable usage outdoors. Additionally, using lightweight yet strong materials such as carbon fiber or aluminum can make installation easier while improving durability and efficiency. This approach allows easy scalability and quick installation across large public areas such as sidewalks, railway stations, school corridors, and parks.

CHAPTER 8

REFERENCES

Below is a well-formatted reference list suitable for academic project reports:

1. S. Priya, D. J. Inman, *Energy Harvesting Technologies*, Springer, 2009. https://link.springer.com/book/10.1007/978-0-387-76464-1
2. D. Shen, J. Shu, "Piezoelectric Energy Harvesting Technologies," *IEEE Transactions on Ultrasonics*, 2014.
3. Rohit K., and Mehta P., "Design of Piezoelectric Energy Generation Floor," *International Journal of Engineering Research*, vol. 5, no. 3, 2018. https://www.ijert.org/
4. Arduino Documentation, "Arduino Uno Technical Specifications," Official Arduino Website, 2024. https://docs.arduino.cc/hardware/uno-rev3
5. Espressif Systems, "ESP8266 and ESP32 Technical Reference Manual," 2023.
6. Blynk IoT Platform Documentation, "Cloud Dashboard & API Guide," 2024.
7. S. Savio, "Human Power Generation Using Piezoelectric Materials," *International Journal of Advanced Research in Electrical Engineering*, 2017. https://www.ijareeie.com/
8. V. P. Singh, "Energy Harvesting from Footsteps using Piezoelectric Sensors," *Journal of Renewable Energy Systems*, 2019. https://www.jres.org/
9. A. Kumar, "IoT-Based Monitoring of Small-Scale Energy Harvesting Systems," *IEEE Internet of Things Journal*, 2022. https://ieeexplore.ieee.org/
10. M. Patel, "Smart City Energy Solutions Using Piezoelectric Walkways," *Smart Infrastructure Review*, 2021. *https://www.smartinfrastructurereview.com/*